

PAVAN KALYAN NEMALAPURI

B.Tech, Electronics and Communication Engineering (2023–27) | KL University
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Portfolio: pavankalyannemalapuri-cell.github.io/portfolio



Skills

Programming Languages: Java, SQL, C programming

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming (Java), DBMS

FPGA / Verilog HDL Design: Digital Design, Hardware Description Language (Verilog)

Projects

Automated Greenhouse Management System

Sep 2025

- Addressed inconsistent greenhouse monitoring by engineering an ESP32-based IoT automation system with DHT11 and soil moisture sensors for real-time environmental tracking.
- Eliminated manual irrigation through closed-loop control that automatically operates a water pump, heater, and cooler based on live sensor thresholds.
- Removed the need for on-site checks by building a remote monitoring interface for continuous visibility into environmental conditions and actuator status.

Up/Down Counter using Master-Slave JK Flip-Flops on Artix-7 FPGA

Feb 2026

- Designed and implemented a 3-bit synchronous Up/Down counter using master-slave JK flip-flops in Verilog HDL.
- Deployed the design on an Artix-7 (xc7a35tftg256-1) FPGA and interfaced it with an on-board 4-digit common-anode 7-segment display.
- Implemented direction control via a selection input and integrated clock division for human-visible output.
- Verified functionality through simulation and hardware validation.

Automatic Railway Gate Control System (ESP8266)

May 2025

- Designed and developed an IoT-based automatic railway gate control system using ESP8266, IR sensors, and servo motors to enhance railway crossing safety.
- Built real-time train detection to autonomously control gate operations and prevent accidents.
- Implemented embedded C programming for sensor interfacing and actuator control with low-latency response.
- Ensured reliable operation through efficient signal processing and hardware integration, demonstrating skills in embedded systems, IoT, and microcontroller-based control.

Certifications

- Embedded System Application and IoT Programming
- VLSI Design and Verification

Achievements

- Merit Scholarship, KL University — Awarded Rs. 4.8 Lakhs merit scholarship for academic excellence in Intermediate, with 50% tuition fee concession each semester.

Education

KL University — B.Tech, Electronics and Communication Engineering
CGPA: 9.8

Jul 2023 – May 2027

SR Junior College — Andhra Pradesh Board of Intermediate Education
Percentage: 98.3%

Jun 2021 – Mar 2023

Sai Vineeth School — Board of Secondary Education, Andhra Pradesh
Percentage: 97.5%